Mediator

Season: 3 · Behavioral · Category: GoF · Level: Intermediate · Build with Waqar

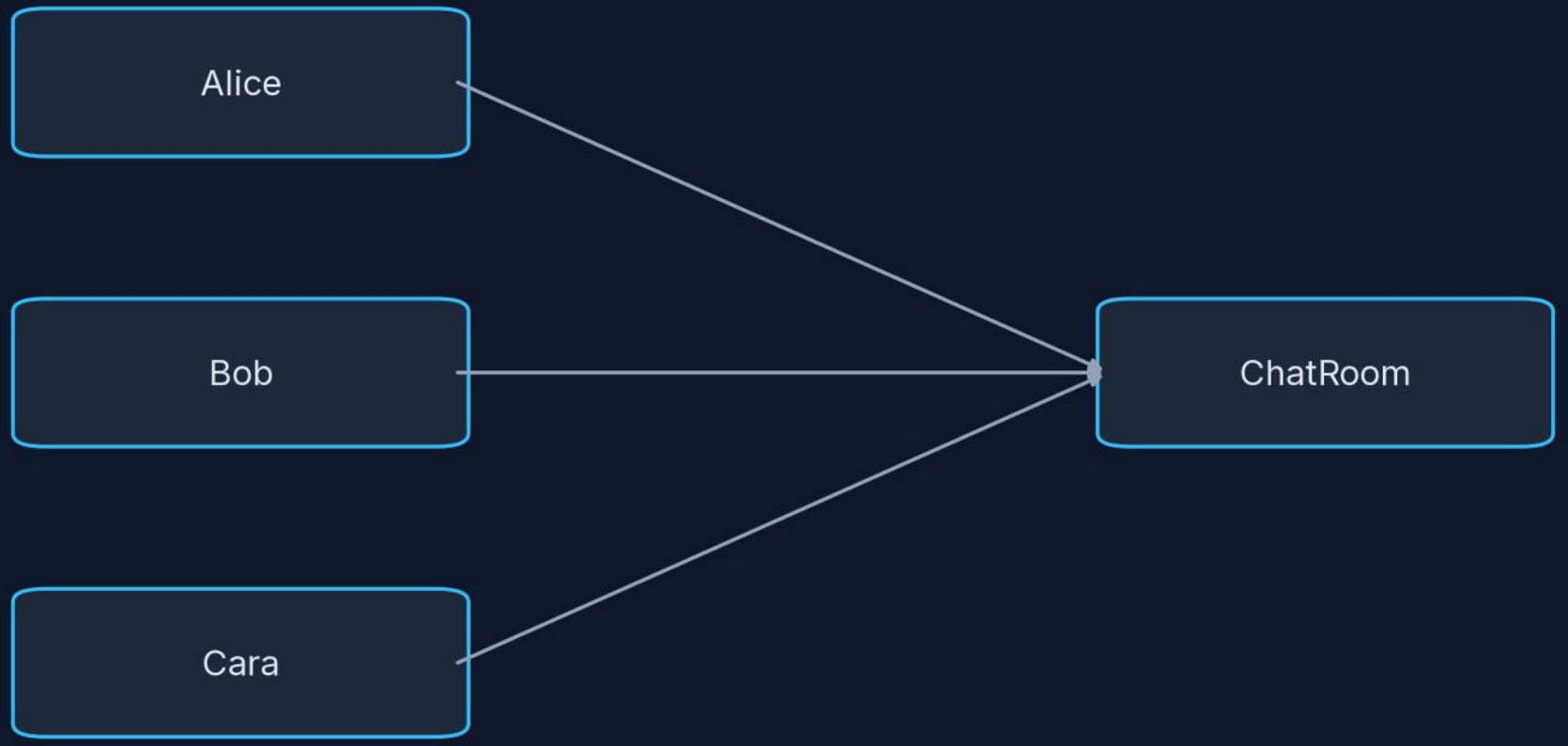
Today's Hook

- Today: Mediator as a StreamForge product force — not trivia.
- Runnable Demo on JDK 25 with captured stdout.
- Classic encoding → Java 25 / Zazr modernization → trade-offs.
- CapCut slides + full spoken script included.

The Problem

- Naive local fixes: more ifs, statics, nulls, or threads.
- Illegal states still compile; tests become order-dependent.
- Design reviews lack a shared name for the force.
- Interviews punish memorization without judgment.

Structure



Mediator structure

Real-World Scenario

- StreamForge creator platform is the running metaphor.
- Same names appear in naive code, refactor, and Demo output.
- Stay in one story for the full 10–12 minutes.
- Product language first; pattern vocabulary second.

Naive Code

```
// Naive sketch for Mediator
public class Naive {
  // duplicated logic / leaked details / unsafe sharing
  Object doIt(Object... args) {
    if (args == null) return null;
    return args[0];
  }
}
```

Why Naive Hurts

- Coupling spreads across controllers and infrastructure.
- Tests need sleeps or static reset hooks.
- New variants require shotgun edits.
- On-call cannot reason about failure modes.

Implementation

```java
// file: src/main/java/com/buildwithwaqar/patterns/gof/behavioral/mediator/ChatRoom.java
package com.buildwithwaqar.patterns.gof.behavioral.mediator;

import java.util.ArrayList;
import java.util.List;

interface ChatMediator {
    void register(User user);
    void send(String from, String message);
}

public final class ChatRoom implements ChatMediator {
    private final List<User> users = new ArrayList<>();
    private final List<String> log = new ArrayList<>();

    public void register(User user) { users.add(user); }

    public void send(String from, String message) {
        var line = from + ": " + message;
        log.add(line);
        for (var u : users) if (!u.name().equals(from)) u.receive(line);
    }
```

Implementation (2)

```
    List<String> log() { return List.copyOf(log); }
}

final class User {
    private final String name;
    private final ChatMediator mediator;
    private final List<String> inbox = new ArrayList<>();

    User(String name, ChatMediator mediator) {
        this.name = name;
        this.mediator = mediator;
        mediator.register(this);
    }

    String name() { return name; }
    void send(String msg) { mediator.send(name, msg); }
    void receive(String msg) { inbox.add(msg); }
    List<String> inbox() { return List.copyOf(inbox); }
}
```

Demo.main

```java
package com.buildwithwaqar.patterns.gof.behavioral.mediator;
public final class Demo {
  public static void main(String[] a) {
    System.out.println("=== Mediator walkthrough ===");
    var room = new ChatRoom();
    var alice = new User("alice", room);
    var bob = new User("bob", room);
    var cara = new User("cara", room);
    alice.send("stand-up in 5");
    bob.send("ack");
    System.out.println("bobInbox=" + bob.inbox());
    System.out.println("caraInbox=" + cara.inbox());
    System.out.println("aliceInbox=" + alice.inbox());
    System.out.println("log=" + room.log());
  }
}
```

Walkthrough Output

```
=== Mediator walkthrough ===
bobInbox=[alice: stand-up in 5]
caraInbox=[alice: stand-up in 5, bob: ack]
aliceInbox=[bob: ack]
log=[alice: stand-up in 5, bob: ack]
```

Step-by-Step Trace

- Each stdout line maps to a refactor stage.
- Re-run Demo locally on JDK 25 to verify.
- Differences usually mean a dirty tree — pull main.
- Pause the video and predict the next line before revealing.

Modern Java 25 / Zazr

```
// Java 25 / Zazr direction for Mediator
// records + sealed + switch
// Option/Either/Validation instead of null
// virtual threads for blocking fan-out
// inject interfaces at the edge
```

Trade-offs

- Clarity vs ceremony — do not spawn five types for two lines.
- Closed sealed worlds vs open SPI plugins.
- Teaching demos omit production idempotency/metrics.
- Prefer simpler language features when they erase the force.

Related Patterns

- Contrast with the neighbor pattern (force difference).
- Name the roles from memory in sixty seconds.
- Point to the season map for what comes next.
- Link the GitHub episode folder in the description.

Recap

- Force → naive pain → Mediator → Demo stdout.
- Modernize with Java 25 / Zazr when it shrinks code.
- Teach when NOT to use it.
- Practice: break a test, fix it, re-run Demo.